

Value Driven Drug Development

Customer Centric model for guiding strategy, managing execution, and delivering the right solutions for providers, practitioners, and patients.

Value Driven Lifecycle



- ☐ Stakeholder Assessment
 - ☐ Business Needs
 - ☐ Compliance Needs
 - ☐ Customer Needs
- ☐ Continuous Improvement
 - ☐ Issues
 - ☐ Opportunities
- ☐ Research
 - ☐ Product / Services Pipeline
 - ☐ Innovative Materials & Science
- ☐ Risk Mitigation
 - ☐ Risk Profile Gaps
 - ☐ Competitive Intelligence

Architecture

Methodologies

Models

Measures

Standards

Value Driven Lifecycle (Discover)



- ☐ Stakeholder Assessment
 - ☐ Business Needs
 - ☐ Compliance Needs
 - ☐ Customer Needs
- ☐ Continuous Improvement
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Architecture:

Business, Information

Methodologies:

EPM, PLM, ECM

Models:

Business, Product, Service, Risk, ...

Measures:

Quality, Financial, Productivity, Value

Standards:

Regulatory, Technical, Operational, ...

Value Driven Lifecycle (Discover)



- ☐ Requirements Engineering
 - ☐ Scenarios, Use Cases, Features
 - ☐ Process Models, Rules, Constraints
 - ☐ Measures
- ☐ Scope & Boundaries
 - ☐ Desired Outcomes, Capabilities
 - ☐ Product & Service Portfolio
 - ☐ Value Platforms, LOB Extensions
- ☐ Financial Commitment
 - ☐ Budget (cost)
 - ☐ Investment Analysis (return)
- ☐ Compliance
 - ☐ Regulatory, Legal, Standards

Architecture:

Business, Information

Methodologies:

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Models:

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Measures:

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Value Driven Lifecycle (Discover)



- ☐ Automation & Interoperability
 - ☐ Data Capture & On Demand Insights
 - ☐ Services, Data & Process Integration
- ☐ Operational Excellence
 - ☐ Transition Products into Operation
 - ☐ Batch Record & Process Plan
 - ☐ Procurement & Capacity Planning
- ☐ Alignment with Customer Needs
 - ☐ Patients, Practitioners, Providers
 - ☐ Operations, Supply Chain Partners
- ☐ Compliance & Quality
 - ☐ Regulatory, Legal, Standards
 - ☐ Data, Process, Systems

Architecture:

Information, Application, Operational

Methodologies:

EPM, PLM, SCM

Models:

Product, E-BOM, M--BOM

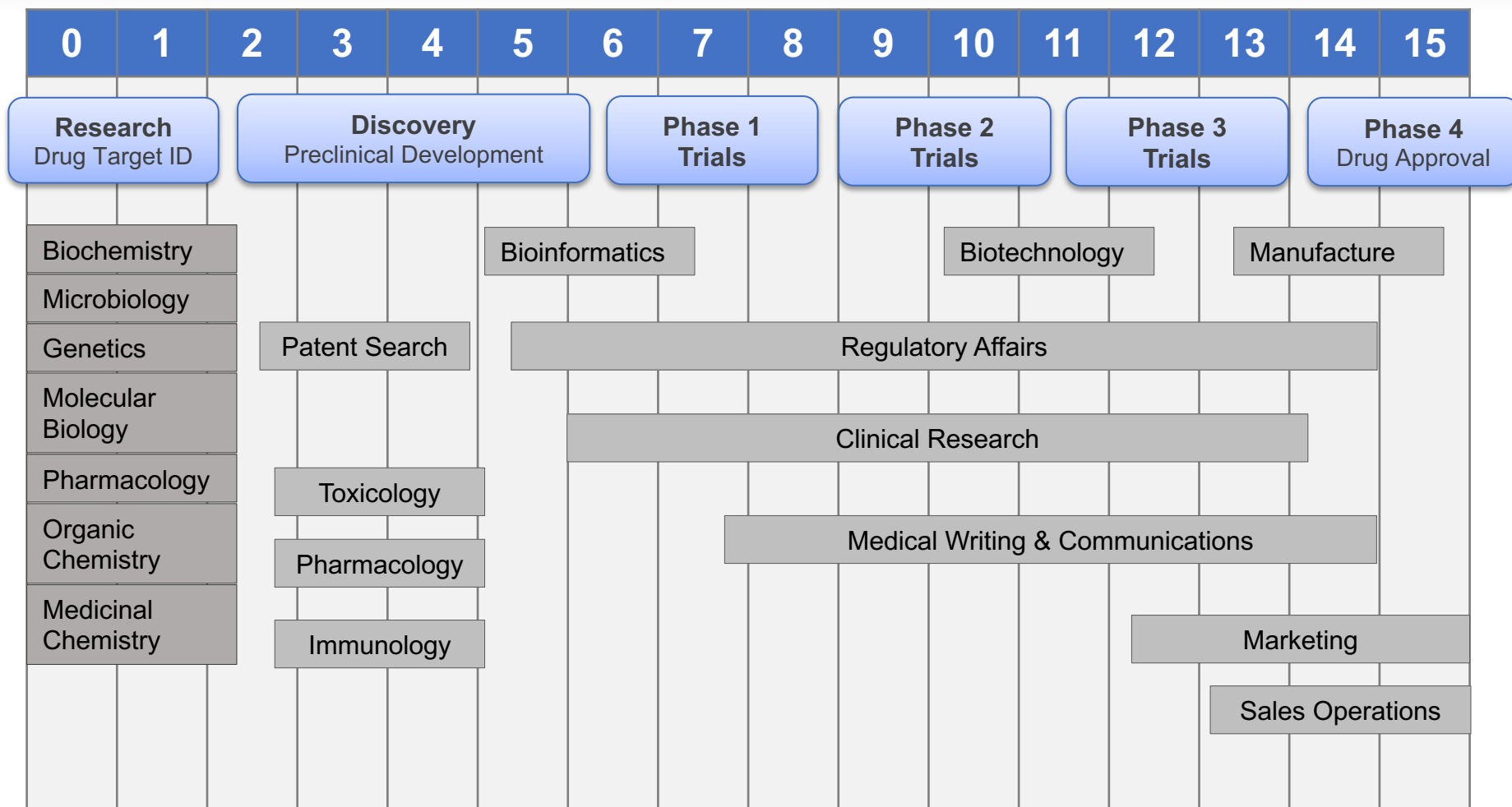
Measures:

Quality, Financial, Productivity, Value

Standards:

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New Drug Delivery Timeline (Template)



Value Driven Lifecycle

(Clinical through Submission)



- ☐ Stage 1 – Pipeline
- ☐ Stage 2 – Research
- ☐ Stage 3 – Preclinical
- ☐ Stage 4 – Phase 1
- ☐ Stage 5 – Phase 2
- ☐ Stage 6 – Phase 3
- ☐ Stage 7 – Phase 4
- ☐ Stage 8 - Submission

Key Decisions after each stage

Pipeline:

- Customer Focus (new / existing)
- Identify Candidate Drugs
- Screen for feasibility & viability
- Assess Market Opportunity
- Define critical project issues and draft the plan & schedule
- Outline financial plan & budget

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(Clinical through Submission)



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Key Decisions after each stage

Research:

- IP Search and Patent exploration
- Optimize drug candidate (chem)
- Early DMPK (drug metabolism & pharmacokinetics)
- Initial formulation
- Pharmacology
- Optimize IP
- Validate the market potential
- Issue & Risk Management
- Draft WBS for Schedule (incl. dep)
- Include Regulatory Strategy in plan

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Key Decisions after each stage

Preclinical:

- Initiate Special Pharmacology
- Initiate Toxicology & Pathology
- Begin pharma development
- Scale up drug supply
- Conduct DPMK
- Submit IND (investigational)
- Plan Clinical Development

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Key Decisions after each stage

Phase 1:

- Conduct Clinical Protocol studies
- Conduct pK, pD interaction studies
- Target product profile (initial)
- Draft detailed plan for Phases

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(Clinical through Submission)



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Key Decisions after each stage

Phase 2:

- Continue / Complete Clinical Protocol studies
- Continue / Complete pK, pD interaction studies
- Draft Target product profile (detailed)
- Scale up validation
- Update Market Analysis (Quant)
- Update detailed plan for later Phases

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(Clinical through Submission)



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Key Decisions after each stage

Phase 3:

- Continue / Complete Clinical Protocol studies
- Define / Refine dosage
- Conduct efficacy & safety
- Clarify CTD (ISE / ISS)
- Define CMC (chemistry, mfg., & controls) section for IND
- Validate production scale
- Validate production line, sourcing, and manufacturing capability
- Refine detailed plan for later Phases

Value Driven Lifecycle

(Clinical through Submission)



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Key Decisions after each stage

Phase 4:

- Complete Clinical Protocol studies
- Clarify dosage, form, strength
- Complete efficacy & safety
- Update CTD (ISE / ISS)
- Update CMC (chemistry, mfg., & controls) section for IND
- Initiate pre-launch marketing
- Draft publications
- Initiate Dossier assembly
- Draft schedule for submission

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Key Decisions after each stage

Submission:

- Prepare Dossier
- Prepare symposia and publications
- Draft detailed marketing plan
- Initiate Marketing activities
- Draft Transition Plan (R&D to Operations)
- Draft vendor (supplier) contracts
- Draft operation plans
- Draft quality plans
- Prepare NDA
- Review NDA
- Submit NDA & Manage responses from regulatory agencies

Measures Model

Trial Progress / Timeliness Metrics

Investigator recruitment and startup
 Patient recruitment / enrollment
 CRF (Case Rpt. Form) Progress
 Clinical supply chain visibility
 Milestones on-target / missed
 Contract signed on-time
 Protocol issues on-time
 FPFV (first patient first visit) on-time
 Standards compliance

Queries / CRF Resolutions (Type 1 & 2)
 Adverse Events
 Patient Satisfaction
 Physical satisfaction
 Investigator and site performance
 Error Rates in CDMS (clinical data mgmt.)
 LPLV (last patient last visit) to DB lock
 DB Lock to final results

Quality Metrics

Financial Metrics

Earned Value
 Capability Throughput
 Actual Forecast
 Pipeline
 Pre-proposal financials
 Patient recruitment, advertising and incentives
 Average per-patient costs by trial / therapeutic area
 Cycle times and time-based measures
 Resource allocation-based measures
 Efficiency and operational effectiveness

Staff performance by site
 Staff / contractor retention
 Timeliness of staff through all phases
 Programming productivity (complexity)
 Forecast loading for skilled resources
 Enrollment quotas
 EDC (electronic data capture) performance
 Patients per CRA (clinical research associate)

Resource Metrics

Common Goals for Optimizing Drug Development

Current State

- Disconnected interfaces
- Manual processes and interventions
- Limited standardized processes
- Lot of customization
- Rework
- Variable quality
- Wait time
- Missing information
- High costs
- Long cycle times
- Duplication of efforts

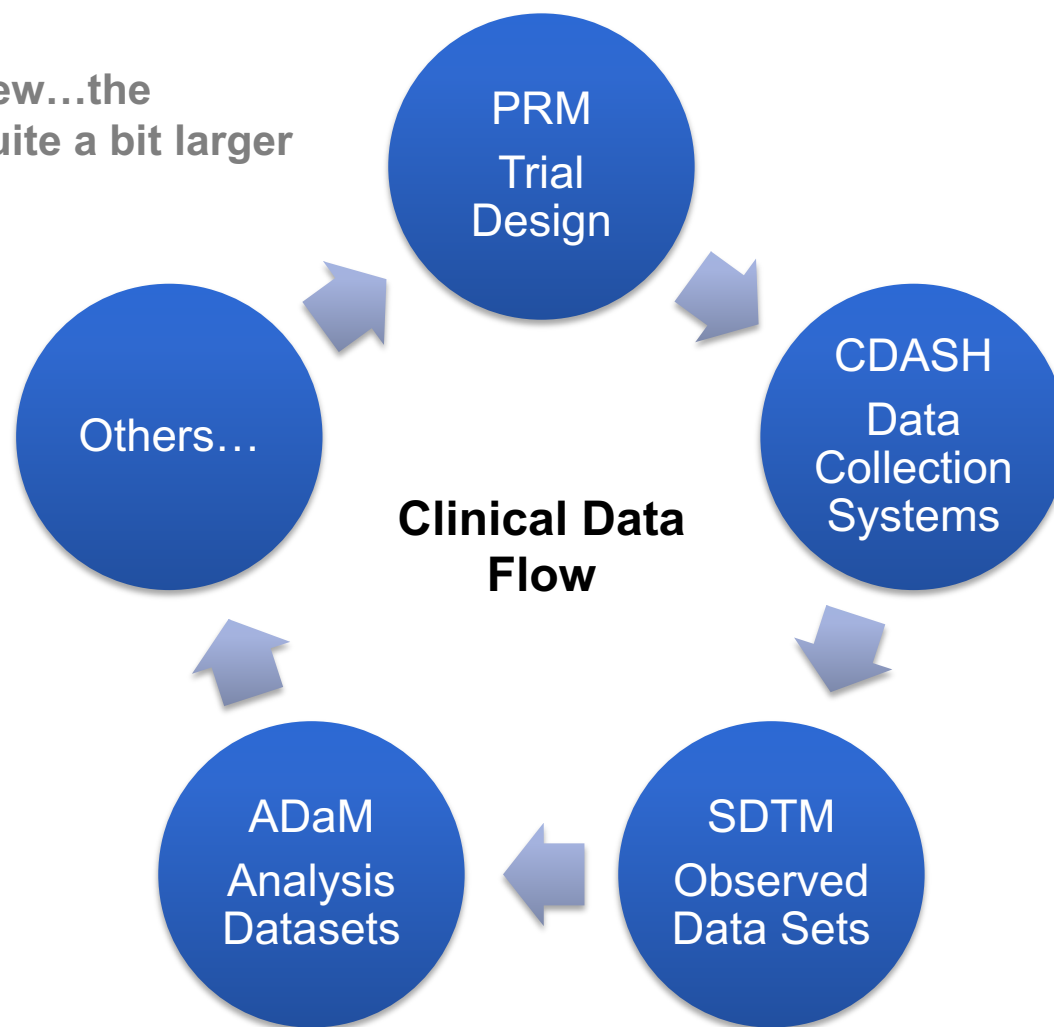
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- ☐ Data Integrity
 - ☐ Risk Profiling
 - ☐ Collaboration
 - ☐ End-2-End Analytics
 - ☐ Impact Analysis
 - ☐ Common Protocols

Future State

- Patient-centric clinical trial design
- End-to-end electronic data flow
- Seamless interfaces
- Improve Automation
- Create Transparency
- Increased standardized processes
- Reduce rework
- Quality across the lifecycle
- Reduce Cycle Time
- Cost efficient
- Integrate Regulatory, Safety and Medical Sciences
- Eliminate redundancies
- Parallel clinical trials
- Extended Data Analytics

Emerging Industry Standards to consider

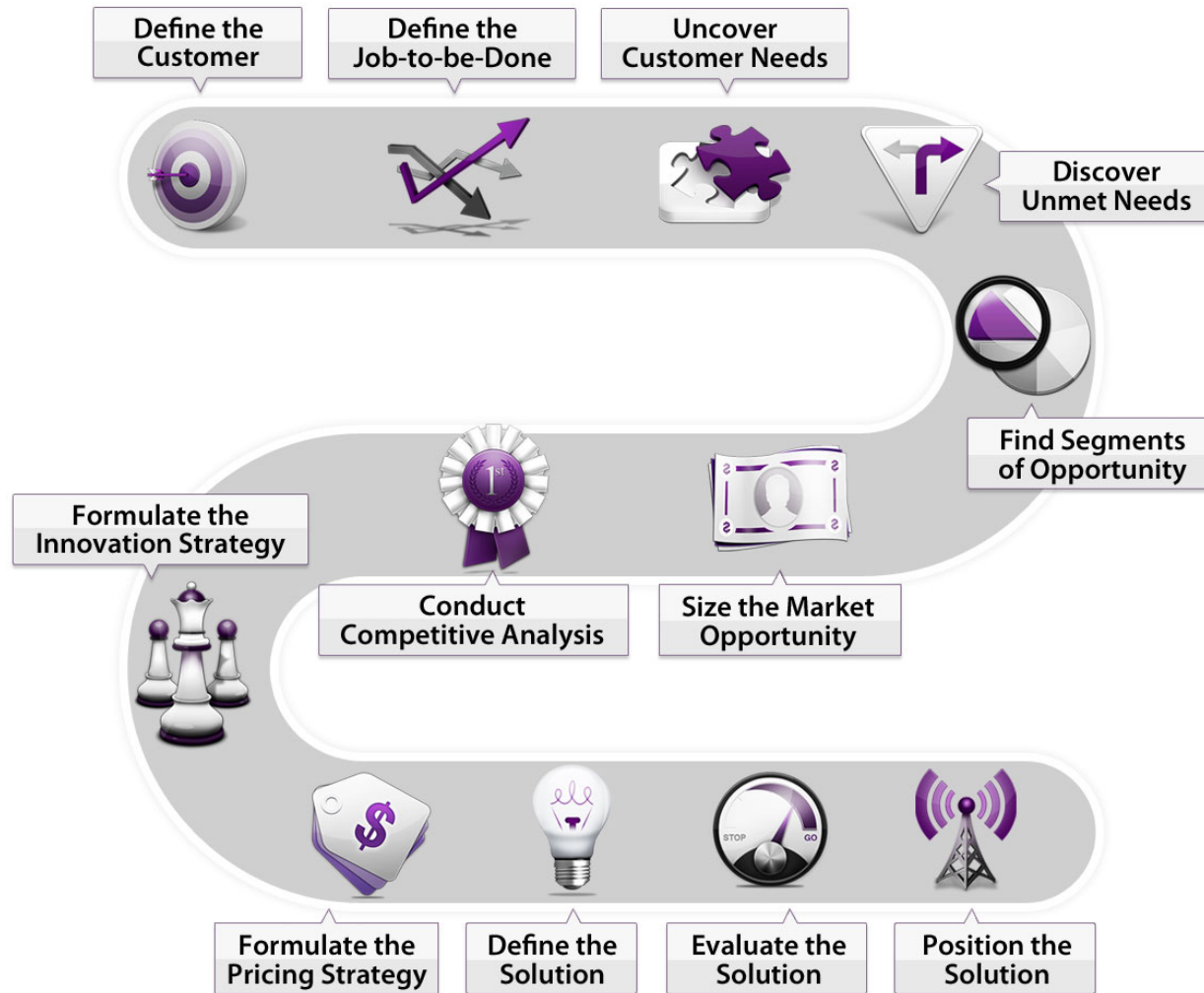
These are just a few...the complete list is quite a bit larger



Customer Journey Mapping

GUIDING VALUE CREATION & DELIVERY

Understanding Customer Needs



Customer journey mapping

- ❑ Customer journey mapping is the process of tracking and describing all the experiences that customers have as they encounter a service or set of services, taking into account not only what happens to them, but also their responses to their experiences.
- ❑ Used well, it can reveal opportunities for improvement and innovation in that experience, acting as a strategic tool to ensure every interaction with the customer is as positive as it can be.

Customer journey mapping

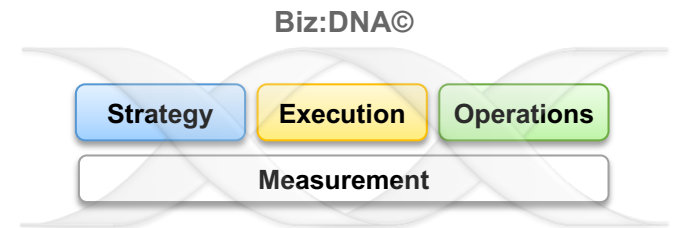
- ❑ It is a qualitative approach, focused on emotional insights about a customer, in order to tell his or her story with passion and narrative. It's a powerful way of engaging both staff and customers.
- ❑ Mapping the system, or process mapping, maps the steps in a process and identifies where to act to make the experience as easy, pleasant and efficient as possible.
- ❑ Measuring the experience is a form of mapping that allows you to determine how well an experience is delivered. It can quantify the effect of changes and contribute to business cases.

Value of Customer journey mapping

- ❑ Journey mapping can deliver a better customer experience and bring about greater efficiencies. More satisfied customers are cheaper to serve and easier to deal with. Good customer understanding is also good business.

Thank You

Measure what matters
Create and deliver the right value
Build and nurture a winning culture
Proactively manage change & complexity



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